

Student Guide — senior / module-11- multiplication-near-base

IKS Curriculum

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Student Workbook

Student Workbook — Multiplication Near a Power of 10 — Nikhilam (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Nikhilam* — _____
- *vinkulam* — _____
- *ādhikya* — _____
- *Ūrdhva-tiryagbhyām* — _____

Day-by-day notes

Day 1 — Finger-Trick Prelude

What I learned:

One question I have:

Day 2 — Two-Digit $\times$ Two-Digit Near 100 (Below)

What I learned:

One question I have:

Day 3 — Two-Digit $\times$ Two-Digit Near 100 (Above)

What I learned:

One question I have:

Day 4 — Mixed — One Above, One Below

What I learned:

One question I have:

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Day 7 — When To Use It

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Day 8 — Speed Sprints

What I learned:

One question I have:

Day 9 — Mixed Practice + Module 7 Crossover

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Finger-Trick Prelude

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: practice the below-10 finger method (e.g., 7×8 using deficits).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Recall Module 9.** Nikhilam = “all from 9, last from 10” — the complement to a base. Same sūtra, NEW application: multiplication.
2. **The below-10 finger demo.** Each hand has 5 fingers; together 10. To multiply 7×8 :
 - Curl $7-5=2$ fingers on left hand.
 - Curl $8-5=3$ fingers on right hand.
 - Curled fingers in total: $5 \rightarrow$ that’s the tens place: 50.
 - Uncurled fingers (left $\times$ right): $3 \times 2 = 6 \rightarrow$ that’s the units: 6.
 - Answer: 56. ✓
3. **Why this works.** Algebraically: $(10 - a)(10 - b) = 100 - 10(a+b) + ab = 10(10-(a+b)) + ab$. The “curled fingers” count is $(a+b)-5$ — wait, let me re-derive. Actually: curled = $(10-a-5) + (10-b-5)$ wait... let me restate cleanly. For 7×8 : $7 = 5+2$ (uncurled fingers

on left), $8 = 5+3$ (right). The formula: tens digit = uncurled-left + uncurled-right – 5; units = uncurled-left $\times$ uncurled-right. Let’s just use the Nikhilam algebra in next lesson.

4. **What we’ll learn this module.** A general method: when factors are CLOSE to a base (10, 100, 1000), Nikhilam-multiplication is much faster than long multiplication.
5. **Speed test.** Compute 8×9 mentally. Compare your method with traditional times table. Most students will use memorisation — the next lessons show why Nikhilam is faster for LARGER near-base products.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Finger-Trick Prelude.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Two-Digit $\times$ Two-Digit Near 100 (Below)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 97×96 using deficits.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. The two-step rule for both factors below 100.

- Find deficits from 100: e.g., $97 \rightarrow -3$, $96 \rightarrow -4$.
- **Left part:** cross-subtract. $97 - 4 = 93$ (or equivalently $96 - 3 = 93$).
- **Right part:** product of deficits. $3 \times 4 = 12$.
- Combine: **9312**.

2. Worked examples.

- $97 \times 96 = 93 \mid 12 = \mathbf{9312}$.
- $88 \times 95 = 83 \mid 60 = \mathbf{8360}$. (Deficits: 12, 5. Cross: $88 - 5 = 83$ or $95 - 12 = 83$. Product: $12 \times 5 = 60$.)
- $99 \times 99 = 98 \mid 01 = \mathbf{9801}$. (Deficits 1, 1. Cross: 98. Product: 1. Pad to 2 digits: 01.)

3. Why the padding matters.

The right part must have as many digits as the base has zeros (here, 2 zeros $\rightarrow$ 2-digit right part). If product is single-digit, pad with leading zero. If product overflows (3+ digits), carry to left part.

4. Algebra.

Base $B = 100$. Factor 1 = $B - a$, factor 2 = $B - b$. Then $(B-a)(B-b) = B(B-a-b) + ab = B \cdot (\text{cross}) + (\text{right})$. The cross-subtract IS $B - a - b$. The product of deficits IS ab .

5. Independent practice.

8 problems, all below-100 case. 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two-Digit $\times$ Two-Digit Near 100 (Below).

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Two-Digit $\times$ Two-Digit Near 100 (Above)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 103×104 using excesses.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: *ādhikya*; lit. “excess (number above base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Above-100 variant.** Use *excesses* instead of deficits.
 - $103 = 100 + 3 \rightarrow$ excess +3.
 - $104 = 100 + 4 \rightarrow$ excess +4.
2. **The two-step rule for above-base.**
 - **Left part:** cross-add. $103 + 4 = 107$ (or $104 + 3 = 107$).
 - **Right part:** product of excesses. $3 \times 4 = 12$.
 - Combine: **10712**.
3. **Worked examples.**

- $103 \times 104 = 107 \mid 12 = \mathbf{10712}$.
 - $110 \times 105 = 115 \mid 50 = \mathbf{11550}$.
 - $102 \times 109 = 111 \mid 18 = \mathbf{11118}$.
4. **Algebra (above-base).** $(B+a)(B+b) = B(B+a+b) + ab$. Cross-ADD = $B+a+b$. Product of excesses = ab .
 5. **Symmetry recognition.** Below-base: cross-SUBTRACT. Above-base: cross-ADD. Both: product-of-difference-from-base. *“This symmetry hints at the general pattern — tomorrow we’ll see the mixed case.”*

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two-Digit $\times$ Two-Digit Near 100 (Above).

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Mixed — One Above, One Below

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 102×98 with sign care.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyām* (IAST: Ūrdhva-tiryagbhyām; lit. ““vertically and crosswise” — the general-purpose sūtra”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed case (one above, one below).** Care with signs! Example: 102×98 .
 - $102 \rightarrow +2$ (excess), $98 \rightarrow -2$ (deficit).
 - Left part: $102 - 2 = 100$ (or $98 + 2 = 100$).
 - Right part: $(+2)(-2) = -4$.
 - Combine: $100 \mid -4$. But how to interpret -4 in the right slot?
2. **The “borrow back” trick.** Negative right part $\rightarrow$ subtract from left. $100 \mid -4$ means $10000 - 4 = 9996$.
3. **Alternative algebra check.** $102 \times 98 = (100+2)(100-2) = 100^2 - 4 = 9996$. ✓
(Difference of squares identity.)
4. **General mixed-case rule.** Cross-subtract (using the deficit sign) for left; multiply excess and deficit (negative result) for right; convert negative right to a borrow from left.
5. **Independent practice.** 5 problems mixing all three cases (both below, both above, mixed). 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Mixed — One Above, One Below.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — 3-Digit $\times$ 3-Digit Near 1000

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 997×996 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Extension to base 1000.** Same rule, base now 1000.
 - $997 \rightarrow$ deficit 3.
 - $996 \rightarrow$ deficit 4.
 - Left: $997 - 4 = 993$.
 - Right: $3 \times 4 = 12$. Must be 3 digits (because base has 3 zeros). Pad: **012**.
 - Combine: **993012**.
2. **Above 1000.** $1003 \times 1004 = 1007 \mid 012 = \mathbf{1007012}$.
3. **Mixed case.** $1003 \times 997 = (1000+3)(1000-3) = 10^6 - 9 = 999991$. (Difference of squares.)

4. **Pattern recognition.** The base extends naturally: 10, 100, 1000, 10000... Each next-decade-base just shifts everything. The CORE rule (cross-subtract + product-of-deficits) is identical.
5. **Independent practice.** 5 near-1000 problems. The challenge: keeping the right-part digit count correct.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces 3-Digit $\times$ 3-Digit Near 1000.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive $(b-a)(b-b) = b(b-a-b) + ab$.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: *vinkulam*; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Derive the formula.** Let B = the base (10, 100, 1000). Factor 1 = $B - a$, Factor 2 = $B - b$.
$$(B - a)(B - b) = B^2 - Ba - Bb + ab = B(B - a - b) + ab = B \cdot (\text{left part}) + (\text{right part})$$

So the left part is $B - a - b = (B - a) - b = (B - b) - a =$ either factor minus the OTHER’s deficit. The right part is just ab .
2. **Why “cross-subtract”?** $B - a - b =$ first factor $- (B - \text{second factor}) =$ first factor $-$ second’s deficit. The cross-subtraction is just rearranging the algebra into a mental-math procedure.
3. **Above-base proof.** $(B + a)(B + b) = B^2 + Ba + Bb + ab = B(B + a + b) + ab$. Cross-add, product of excesses.
4. **Mixed case proof.** $(B + a)(B - b) = B^2 - Bb + Ba - ab = B(B + a - b) - ab$. The “ $-ab$ ” in the right slot is what causes the sign issue.
5. **Connecting the dots.** This is just basic high-school algebra — no mystery. Tirthaji’s sūtra system gives the procedure a memorable name; the algebra gives it the proof.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — When To Use It

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify problems where nikhilam is faster than long multiplication.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: ādhikya; lit. “excess (number above base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The decision tree.**
 - Both factors within ~5 of the same base → Nikhilam is fast.
 - One factor close to base, other far → traditional multiplication often faster.
 - Both factors far from any base (e.g. 47×63) → traditional.
 - Both factors near DIFFERENT bases (e.g. 98×1005) → tricky; pick a common base and scale.
2. **Speed comparison.** Time 97×96 vs 47×63 mentally. Most students will find Nikhilam dominates the first; long multiplication dominates the second.
3. **Pedagogical limit.** Nikhilam multiplication is NOT a universal speed-up. It’s a tool for ONE problem class. Recognising when to use it is the skill.

4. **Practice with strategic choice.** 10 problems, mixed. Each student picks the best method and notes it next to the answer.
5. **Reflection.** “*What pattern of problems makes Nikhilam shine? What pattern defeats it?*” Each student writes 3 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces When To Use It.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Speed Sprints

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: race traditional vs nikhilam on 30 problems.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyām* (IAST: Ūrdhva-tiryagbhyām; lit. ““vertically and crosswise” — the general-purpose sūtra”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Sprint A (Nikhilam-favourable, 10 problems, 4 min).** All factors within 5 of 100 or 1000.
2. **Sprint B (mixed, 10 problems, 4 min).** Mix of near-base and far-from-base.
3. **Sprint C (long-mult-favourable, 10 problems, 4 min).** All factors far from base.
4. **Analysis.** Each student records times. Most: $A < B < C$ dramatically. The CONCRETE evidence that Nikhilam works for ONE class.
5. **Discussion.** “*What’s your strategy choice now? When will you use Nikhilam in real life?*” Each student writes 2 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Sprints.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Mixed Practice + Module 7 Crossover

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve problems combining $\times 11$ and near-base methods.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Crossover with Module 7.** Each student solves 5 problems combining $\times 11$ method + Nikhilam:
 - $11 \times 99 = 1089$ (use $\times 11$)
 - $97 \times 96 = 9312$ (use Nikhilam)
 - $11 \times 98 = 1078$ (use $\times 11$)
 - $105 \times 95 = 9975$ (use difference-of-squares or Nikhilam mixed)
 - $11 \times 87 = 957$ (use $\times 11$ with carry)
2. **Strategy reflection.** Pair-discuss: “*For each, what method DID you use? Was it the fastest?*”
3. **Build a personal toolkit.** Each student updates their “personal mental-math toolkit” page (started in Module 7) to include Nikhilam multiplication.
4. **One more example.** Solve 998×997 mentally. (Nikhilam: deficits 2, 3 $\rightarrow$ left = 995, right = 006 $\rightarrow$ **995006**.) 4 seconds with the trick; >1 minute by long multiplication.
5. **Reflection.** “*What’s the most useful thing you learned this module?*”

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Mixed Practice + Module 7 Crossover.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: summative quiz and method-comparison reflection.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: *vinkulam*; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Method-comparison written reflection (10 min).** “*In 300 words, compare Tirthaji’s near-base multiplication with traditional long multiplication. When does each excel? What does Tirthaji’s method teach about number sense?*”

2. **Summative quiz (25 min).** Mix of Nikhilam (below, above, mixed), algebraic proof, strategy choice. See `quizzes/summative.md`.
3. **Personal mental-math toolkit share (10 min).** Each student shows their one-page toolkit.
4. **Module close.** “*One number-sense habit you’ll carry forward?*”
5. **Linkage.** Module 14 (Magic Squares) will use related place-value insights for grid arithmetic.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Multiplication Near a Power of 10 — Nikhilam (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Multiplication Near a Power of 10 — Nikhilam to a friend who hasn’t taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.

3. Compare Multiplication Near a Power of 10 — Nikhilam with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Multiplication Near a Power of 10 — Nikhilam? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are *doṣas* a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Finger-Trick Prelude** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Two-Digit × Two-Digit Near 100 (Below)** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Two-Digit × Two-Digit Near 100 (Above)** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Mixed — One Above, One Below** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — 3-Digit × 3-Digit Near 1000** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — When To Use It** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Sprints** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.